

## OS-LAB 1

**Write a C program to simulate the following CPU scheduling algorithms to find the average turnaround time and waiting time.**

**a) FCFS**

**CODE:**

```
#include <stdio.h>
#include <stdlib.h>
#include <stdbool.h>
#define n 5

int ct_1=0;
float tat_1=0,wt_1=0;

typedef struct{
    int bt;
    int at;
    int ct;
    int tat;
    int wt;
}process;

process a[n]={0};

int main(){

for(int i=0;i<n;i++){
    printf("enter at, bt for process %d \n",i+1);
    scanf("%d",&a[i].at);
    scanf("%d",&a[i].bt);
```

```
}
```

```
for(int i=0;i<n-1;i++){  
    for(int j=i+1;j<n;j++){  
        if(a[i].at > a[j].at){  
            process temp=a[i];  
            a[i]=a[j];  
            a[j]=temp;  
        }  
    }  
}
```

```
for(int i=0;i<n;i++){  
  
    if(ct_1 < a[i].at)  
        ct_1 = a[i].at;  
  
    ct_1 += a[i].bt;  
  
    a[i].ct=ct_1;  
    a[i].tat=a[i].ct-a[i].at;  
    a[i].wt=a[i].tat-a[i].bt;  
  
    tat_1+=a[i].tat;  
    wt_1+=a[i].wt;  
}
```

```
printf("\nProcess\tAT\tBT\tCT\tTAT\tWT\n");
```

```
for(int i=0;i<n;i++){  
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
```

```

        i+1,a[i].at,a[i].bt,a[i].ct,a[i].tat,a[i].wt);
    }

    printf("average tat=%f,avg wt=%f",tat_1/n,wt_1/n);

}

```

#### OUTPUT:

```

enter at, bt for process 1
0 7
enter at, bt for process 2
8 3
enter at, bt for process 3
3 4
enter at, bt for process 4
5 6

Process AT      BT      CT      TAT      WT
P1      0       7       7       7       0
P2      3       4      11       8       4
P3      5       6      17      12       6
P4      8       3      20      12       9
average tat=9.750000,avg wt=4.750000

```

**OBSERVATION BOOK:**

Write A Program to simulate the following CPU scheduling algorithm

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
```

```
typedef struct {
    int bt;
    int at;
    int ct;
```

```
} process;
```

```
typedef struct {
    int tat;
    int wt;
```

```
} results;
```

```
process a[4] = {0};
results res[4] = {0};
```

```
int main() {
```

```
    for (int i=0; i<4; i++) {
```

```
        // bt, ct
        printf("Enter process times for process %d\n", i+1);
        scanf("%d", &a[i].at);
        scanf("%d", &a[i].bt);
        scanf("%d", &a[i].ct);
```

```
    }
    for (int i=0; i<4; i++) {
```

```
        for (int j=0; j<4; j++) {
```

```
            if (a[i].at > a[j].at) {
```

```
                process temp = a[i];
```

```
                a[i] = a[j];
```

```
                a[j] = temp; } } }
```

```
for(int i=0; i<4; i++) {
```

```
    arr[i].tat = a[i].ct - a[i].at;
```

```
    arr[i].wt = a[i].ct - a[i].at - a[i].bt;
```

```
}
```

```
float sum_tat = 0;
```

```
float sum_wt = 0;
```

```
for(int i=0; i<4; i++) {
```

```
    sum_tat = sum_tat + arr[i].tat;
```

```
    sum_wt = sum_wt + arr[i].wt;
```

```
}
```

```
sum_tat /= 4;
```

```
sum_wt /= 4;
```

```
printf("Avg. TAT is %.1f and Avg. WT is %.1f", sum_tat, sum_wt);
```

```
return 0;
```

```
}
```

Output:

Enter process times for process 1:

0 7 7

Enter process times for process 2:

8 3 20

Enter process times for process 3:

3 4 11

Enter process times for process 4:

5 6 17

Average TAT is 9.75 and average WT is 4.75